Chunyang Luo

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Research Interests: Agent Systems | Human-Agent Interaction | LLM-based Planning & Reasoning | Multi-Robot Planning

EDUCATION

Peking University (PKU) Beijing, China

BEng. in Robotics Engineering 09/2023 – Present (06/2027)

Tentative GPA: **3.54/4.0**

ONGOING MANUSCRIPT

- Zhang, B., Li, J., **Luo, C. (third author)**, Tang, Y., Zhang, W., Piao, S., Liu, D., Ma, S., & Ma, X. (2026). MemoSync: Co-managing developer-agent metamemory across contexts. *ACM CHI 2027 (under review).*

AWARDS AND HONORS

- Second Prize, 22nd "Jiang Zehan Cup" Modeling and Computer Applications Competition, PKU, May 2025
- Honorable Mention, MCM/ICM, 2026
- Merit Student and Third-class Scholarship, Peking University 2025–2026 Academic Year

SKILLS

- **Programming & Tools:** Python, MATLAB, Linux, Git, LaTeX
- **Planning & AI:** LTL, PDDL, MILP, Multi-Robot Task Allocation, LLM-based Planning & Reasoning, Prompt Engineering, Reinforcement Learning
- **Robotics:** ROS, Gazebo, RViz, AI2-THOR
- **English:** IELTS Overall 7.0

RESEARCH EXPERIENCES

OpenGRAIL: Integrating LLM and Classical Planning for Multi-Robot Task Planning in Open Worlds 09/2025 – Present

Researcher Lead, supervised by Prof. Zhongkui Li

- Developed an LLM–classical planning framework that translates natural-language instructions into executable multi-robot plans.
- Designed hierarchical task decomposition and MILP-based heterogeneous robot allocation and scheduling under capability, temporal, spatial, and collaboration constraints.
- Conducted preliminary evaluations in household scenarios, demonstrating higher task-execution accuracy and lower makespan and runtime than existing methods.

Human-AI Collaboration for Long-Horizon Agent Systems 07/2026 – 09/2026

Research Assistant, Hong Kong University of Science and Technology (HKUST), supervised by Assoc. Prof. Xiaojuan Ma

- Investigated how AI agents maintain task context, user intent, and reusable knowledge during long-horizon interactions.
- Studied existing agent memory mechanisms and contributed to the conceptual design of memory representations and workflow structures for improving agent reliability.
- Designed evaluation scenarios and comparison experiments to analyze the impact of memory organization strategies on task completion, consistency, and knowledge reuse.

Multi-Robot Collaborative LTL Continuous Planning for Dynamic Environments 07/2025 – Present

Research Assistant, supervised by Prof. Zhongkui Li

- Designed an online task decomposition and replanning framework for heterogeneous multi-robot LTL tasks under dynamic events.
- Built a heterogeneous multi-robot simulation platform (Unitree Go2, Scout) with task execution and state feedback interfaces.
- Benchmarked against static and periodic-replanning baselines on completion rate, planning time, and cumulative cost; extended to physical robots.

SELECTED PROJECTS & TRAINING

RepoFlow: Repository-Level Workflow for AI Coding Agents 03/2026 – Present

Course Project

Designed persistent-context and structured agent-handoff mechanisms for task continuity in long-horizon coding agents.

Alibaba DAMO Academy – Embodied Intelligence Bootcamp 03/2026 – 04/2026

Built a robotic-arm visual-navigation demo integrating perception, control, and task execution.